GEMM Analytical Model - Equations

Auto-generated from PYTHON_FORMULAS and LATEX_SYMBOL_OVERRIDES in gemmanalytics.py.

Output order: execution

row 22 - INPUT_A_BYTES_PER_ELEMENT

$$\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matA})} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(\mathrm{matA})}]$$

row 23 - INPUT_B_BYTES_PER_ELEMENT

$$\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matB})} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(\mathrm{matB})}]$$

row 25 - OUTPUT_BYTES_PER_ELEMENT_AFTER_DOWN_CONVERSION

$$\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{D}\downarrow)} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(\mathrm{matD}\downarrow)}]$$

row 33 - EU_COUNT

$$|\mathrm{EU}| = |\mathrm{XeCore}|_{\mathrm{pXeCU}} \times |\mathrm{EU}|_{\mathrm{pXeCore}} \times |\mathrm{XeCU}|$$

row 102 - TOTAL_L2_SIZE_B_FOR_A_SINGLE_INSTANCE

$$|\mathrm{L2Bytes}| = \mathrm{L2BankSize}_{\mathrm{MB}} \times |\mathrm{L2Banks}|_{\mathrm{pXeCU}} \times 1024 \times 1024$$

row 125 - MAX_POSSIBLE_HBM_BW_FREQ_B_CLK

$$\mathrm{MemBW}_{\mathrm{BpClk}}^{\mathrm{max}} = \frac{\mathrm{MemBW}_{\mathrm{GBps}}^{\mathrm{max}}}{\mathrm{f}_{\mathrm{GHz}}^{(\mathrm{GT})}}$$

row 36 - MMA_MAC_THROUGHPUT_PER_XECORE

$$\tau_{\mathrm{mMACpClk\cdot XeCore}}^{(\mathrm{peak})} = \min(\frac{4}{\frac{\mathrm{FLOOR}(\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matA})} \times 2, 1)}{2}}, \frac{4}{\frac{\mathrm{FLOOR}(\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matB})} \times 2, 1)}{2}}) \times \mathrm{D}_{\mathrm{DPAS}} \times 16 \times |\mathrm{EU}|_{\mathrm{pXeCore}}$$

row 42 - CLKS_PER_DPAS

$$\mathrm{CLKS}_{\mathrm{DPAS}} = \frac{\mathrm{M}_{\mathrm{pThread}} \times \mathrm{K}_{\mathrm{pThread}} \times \mathrm{N}_{\mathrm{pThread}}}{\frac{\tau^{(\mathrm{peak})}_{\mathrm{mMACpClk \cdot XeCore}}}{|\mathrm{EU}|_{\mathrm{pXeCore}}}}$$

row 13 - WAVES

$$\mathrm{N_w aves} = \mathrm{CEILING}(|\mathrm{Tiles}|^{(\mathrm{GPU})}_{\mathrm{N}}, 1) \times \mathrm{CEILING}(|\mathrm{Tiles}|^{(\mathrm{GPU})}_{\mathrm{M}}, 1)$$

row 14 - TG_TILES_IN_N

$$|\mathrm{Tiles}|^{(\mathrm{TG})}_{\mathrm{N}} = \frac{\mathrm{N_d im}}{\mathrm{W}^{(\mathrm{ThreadGroup, realized\ by\ multiple\ MMA\ iterations})}_{\mathrm{element}}}$$

row 15 - TG_TILES_IN_M

$$|\mathrm{Tiles}|^{(\mathrm{TG})}_{\mathrm{M}} = \frac{\mathrm{M_d im}}{\mathrm{H}^{(\mathrm{ThreadGroup})}_{\mathrm{element}}}$$

row 16 - TG_CLUSTER_TILES_IN_N

$$|\mathrm{Tiles}|^{(\mathrm{TG\ Cluster})}_{\mathrm{N}} = \frac{\mathrm{N_d im}}{\mathrm{W}^{(\mathrm{XeCoreCluster})}_{\mathrm{element}}}$$

row 17 - TG_CLUSTER_TILES_IN_M

$$|\mathrm{Tiles}|^{(\mathrm{TG\ Cluster})}_{\mathrm{M}} = \frac{\mathrm{M_d im}}{\mathrm{H}^{(\mathrm{XeCoreCluster})}_{\mathrm{element}}}$$

row 18 - XECU_TILES_IN_N

$$|\mathrm{Tiles}|^{(\mathrm{XeCU})}_{\mathrm{N}} = \frac{\mathrm{N_d im}}{\mathrm{W}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}$$

row 19 - XECU_TILES_IN_M

$$|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{XeCU})} = \frac{\mathrm{M_dim}}{\mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})}}$$

row 20 - GPU_TILES_IN_N

$$|\mathrm{Tiles}|_{\mathrm{N}}^{(\mathrm{GPU})} = \frac{\frac{\mathrm{N_dim}}{\mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})}}}{\mathrm{W}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})}}$$

row 21 - GPU_TILES_IN_M

$$|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{GPU})} = \frac{\frac{\mathrm{M_dim}}{\mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})}}}{\mathrm{H}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})}}$$

row 43 - THREAD_WIDTH_IN_UNITS_OF_ELEMENTS

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{Thread})} = \mathrm{N}_{\mathrm{pThread}}$$

row 44 - THREAD_HEIGHT_IN_UNITS_OF_ELEMENTS

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{Thread})} = \mathrm{M}_{\mathrm{pThread}}$$

row 47 - TG_WIDTH_IN_UNITS_OF_ELEMENT_REALIZED_BY_MULTIPLE_MMA_ITERATIONS

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})} = \mathrm{W}_{\mathrm{thread}}^{(\mathrm{ThreadGroup})} \times \mathrm{W}_{\mathrm{element}}^{(\mathrm{Thread})}$$

row 48 - TG_HEIGHT_IN_UNITS_OF_ELEMENT

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{ThreadGroup})} = \mathrm{H}_{\mathrm{thread}}^{(\mathrm{ThreadGroup})} \times \mathrm{H}_{\mathrm{element}}^{(\mathrm{Thread})}$$

row 52 - XECORE_CLUSTER_WIDTH_IN_UNITS_OF_ELEMENT

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCoreCluster})} = \mathrm{W}_{\mathrm{TG,keep\ cluster\ sizeas4}}^{(\mathrm{XeCoreCluster})} \times \mathrm{W}_{\mathrm{element}}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})}$$

row 53 - XECORE_CLUSTER_HEIGHT_IN_UNITS_OF_ELEMENT

$$\mathrm{H}_{\text{element}}^{(\text{XeCoreCluster})} = \mathrm{H}_{\text{TG,keep cluster sizeas4}}^{(\text{XeCoreCluster})} \times \mathrm{H}_{\text{element}}^{(\text{ThreadGroup})}$$

row 56 - XECU_TILE_WIDTH_IN_UNITS_OF_ELEMENT

$$\mathrm{W}_{\text{element}}^{(\text{XeCUTile})} = \mathrm{W}_{\text{TG}}^{(\text{XECUTile})} \times \mathrm{W}_{\text{element}}^{(\text{ThreadGroup,realized by multiple MMA iterations})}$$

row 57 - XECU_TILE_HEIGHT_IN_UNITS_OF_ELEMENT

$$\mathrm{H}_{\text{element}}^{(\text{XeCUTile})} = \mathrm{H}_{\text{TG}}^{(\text{XECUTile})} \times \mathrm{H}_{\text{element}}^{(\text{ThreadGroup})}$$

row 59 - GPU_TILE_HEIGHT_IN_XECU_UINT

$$\mathrm{H}_{\text{XeCU}}^{(\text{GPUTile})} = \frac{|\text{XeCU}|}{\mathrm{W}_{\text{XeCU}}^{(\text{GPUTile})}}$$

row 60 - GPU_TILE_WIDTH_IN_UNITS_OF_ELEMETNS

$$\mathrm{W}_{\text{element}}^{(\text{GPUTile})} = \mathrm{W}_{\text{XeCU}}^{(\text{GPUTile})} \times \mathrm{W}_{\text{element}}^{(\text{XeCUTile})}$$

row 61 - GPU_TILE_HEIGHT_IN_UNITS_OF_ELEMETNS

$$\mathrm{H}_{\text{element}}^{(\text{GPUTile})} = \mathrm{H}_{\text{XeCU}}^{(\text{GPUTile})} \times \mathrm{H}_{\text{element}}^{(\text{XeCUTile})}$$

row 63 - MAT_A_INPUT_SIZE_B

$$\mathrm{Size}_{\mathrm{B}}^{(\text{matA})} = \mathrm{M}_{\mathrm{d}}\mathrm{im} \times \mathrm{K}_{\mathrm{d}}\mathrm{im} \times \mathrm{Bytes}_{\text{pElement}}^{(\text{matA})}$$

row 64 - MAT_B_INPUT_SIZE_B

$$\mathrm{Size}_{\mathrm{B}}^{(\text{matB})} = \mathrm{K}_{\mathrm{d}}\mathrm{im} \times \mathrm{N}_{\mathrm{d}}\mathrm{im} \times \mathrm{Bytes}_{\text{pElement}}^{(\text{matB})}$$

row 65 - MAT_C_INPUT_D_OUTPUT_SIZE_B

$$\mathrm{Size}_{\text{matB}}^{(\text{matC,matD})} = \mathrm{M}_{\mathrm{d}}\mathrm{im} \times \mathrm{N}_{\mathrm{d}}\mathrm{im} \times \mathrm{Bytes}_{\text{pElement}}^{(\mathrm{D}\downarrow)}$$

row 66 - MAT_D_INTERMEDIATE_SIZE_B

$$\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matD}\downarrow)} = \mathrm{M_dim} \times \mathrm{N_dim} \times \mathrm{Byte}_{\mathrm{pElement}}^{(\mathrm{matD})}$$

row 103 - WORKING_DATA_SET_SIZE_OF_K_IN_L2_CORRESP_20K_CLOCKS_OF_THREAD_DIVERGENCE

$$|\mathrm{WorkingSet}|_{\mathrm{20Kclksofthreaddivergence}}^{(\mathrm{KinL2})} = \min(20000 \times \frac{\mathrm{K_{pThread}}}{\mathrm{CLKS_{DPAS}}}, \mathrm{K_dim})$$

row 104 - TOTAL_REQUIRED_L2_SIZE_FOR_IDEAL_HIT_RATE_B_FOR_A_SINGLE_INSTANCE_AND_SINGLE_WAVE

$$\substack{\mathrm{ance,\ 1\ wave} \\ \mathrm{alwayshit}} = \mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})} \times \mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matA})} \times |\mathrm{WorkingSet}|_{\mathrm{20Kclksofthreaddivergence}}^{(\mathrm{KinL2})} + \mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})} \times \mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matB})} \times |\mathrm{WorkingSet}|_{\mathrm{20Kclksofthreaddivergence}}^{(\mathrm{KinL2})} + \mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})} \times \mathrm{H}$$

row 37 - CLK_SPECIFIED_EFFICIENCY

$$\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})} = \frac{\frac{\frac{\frac{\mathrm{M_dim \times K_dim \times N_dim}}{|\mathrm{XeCore}|_{\mathrm{pXeCU}}}}{|\mathrm{XeCU}|}}{\tau_{\mathrm{mMACpClk \cdot XeCore}}^{(\mathrm{peak})}}}{\eta_{\mathrm{systolic}}}$$

row 69 - TOTAL_L2_READ_B

$$\mathrm{L2Rd}_{\mathrm{B}}^{(\mathrm{total})} = \mathrm{Size}_{\mathrm{B}}^{(\mathrm{matA})} \times \mathrm{CEILING}(\frac{\mathrm{N_dim}}{\mathrm{W}_{\mathrm{element}}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})}}, 1) + \mathrm{Size}_{\mathrm{B}}^{(\mathrm{matB})} \times \mathrm{CEILING}(\frac{\mathrm{M_dim}}{\mathrm{H}_{\mathrm{element}}^{(\mathrm{ThreadGroup})}}, 1)$$

row 70 - TOTAL_L2_WRITE_B

$$\mathrm{L2Wr}_{\mathrm{B}}^{(\mathrm{total})} = \mathrm{Size}_{\mathrm{matB}}^{(\mathrm{matC,matD})}$$

row 71 - TOTAL_L1_READ_B

$$\mathrm{L1Rd}_{\mathrm{B}}^{(\mathrm{total})} = \mathrm{Size}_{\mathrm{B}}^{(\mathrm{matA})} \times \mathrm{CEILING}(\frac{\mathrm{N_dim}}{\mathrm{W}_{\mathrm{element}}^{(\mathrm{Thread})}}, 1) + \mathrm{Size}_{\mathrm{B}}^{(\mathrm{matB})} \times \mathrm{CEILING}(\frac{\mathrm{M_dim}}{\mathrm{H}_{\mathrm{element}}^{(\mathrm{Thread})}}, 1)$$

row 72 - TOTAL_L1_WRITE_B

$$\mathrm{L1Wr}_{\mathrm{B}}^{(\mathrm{total})} = \mathrm{Size}_{\mathrm{matB}}^{(\mathrm{matC,matD})}$$

row 74 - L2_READ_B_XECORE_CLK

$$\mathrm{L2RdB}_{\mathrm{p(Clk \cdot XeCore)}} = \frac{\frac{\mathrm{L2Rd}_{\mathrm{B}}^{\mathrm{(total)}}}{|\mathrm{XeCU}| \times |\mathrm{XeCore}|_{\mathrm{pXeCU}}}}{\mathrm{T}_{\mathrm{clk}}^{\mathrm{(total)}}}$$

row 75 - L2_WRITE_B_XECORE_CLK

$$\mathrm{L2WrB}_{\mathrm{p(Clk \cdot XeCore)}} = \frac{\frac{\mathrm{L2Wr}_{\mathrm{B}}^{\mathrm{(total)}}}{|\mathrm{XeCU}| \times |\mathrm{XeCore}|_{\mathrm{pXeCU}}}}{\mathrm{T}_{\mathrm{clk}}^{\mathrm{(total)}}}$$

row 76 - L2_READ_WRITE_B_XECORE_CLK

$$\mathrm{L2RdWrB}_{\mathrm{p(Clk \cdot XeCore)}} = \mathrm{L2RdB}_{\mathrm{p(Clk \cdot XeCore)}} + \mathrm{L2WrB}_{\mathrm{p(Clk \cdot XeCore)}}$$

row 77 - L1_READ_B_XECORE_CLK

$$\mathrm{L1RdB}_{\mathrm{p(Clk \cdot XeCore)}} = \mathrm{L1RdB}_{\mathrm{p(Clk \cdot EU)}} \times |\mathrm{EU}|_{\mathrm{pXeCore}}$$

row 78 - L1_WRITE_B_XECORECLK

$$\mathrm{L1WrB}_{\mathrm{p(Clk \cdot XeCore)}} = \mathrm{L1WrB}_{\mathrm{p(Clk \cdot EU)}} \times |\mathrm{EU}|_{\mathrm{pXeCore}}$$

row 79 - L1_READ_B_EU_CLK

$$\mathrm{L1RdB}_{\mathrm{p(Clk \cdot EU)}} = \frac{\frac{\mathrm{L1Rd}_{\mathrm{B}}^{\mathrm{(total)}}}{|\mathrm{EU}|}}{\mathrm{T}_{\mathrm{clk}}^{\mathrm{(total)}}}$$

row 80 - L1_WRITE_B_EU_CLK

$$\mathrm{L1WrB}_{\mathrm{p(Clk \cdot EU)}} = \frac{\frac{\mathrm{L1Wr}_{\mathrm{B}}^{\mathrm{(total)}}}{|\mathrm{EU}|}}{\mathrm{T}_{\mathrm{clk}}^{\mathrm{(total)}}}$$

row 83 - L2_READ_MAX_B_XECORE_CLK

$$\mathrm{L2RdB}^{(\mathrm{max})}_{\mathrm{p(Clk\cdot XeCore)}} = \frac{|\mathrm{XeCore}|_{\mathrm{pXeCU}} \times 64}{|\mathrm{XeCore}|_{\mathrm{pXeCU}}}$$

row 84 - L2_WRITE_MAX_B_XECORE_CLK

$$\mathrm{L2WrB}^{(\mathrm{max})}_{\mathrm{p(Clk\cdot XeCore)}} = \mathrm{L2RdB}^{(\mathrm{max})}_{\mathrm{p(Clk\cdot XeCore)}}$$

row 85 - L2_READ_WRITE_MAX_B_XECORE_CLK

$$\mathrm{L2RdWrB}^{(\mathrm{max})}_{\mathrm{p(Clk\cdot XeCore)}} = \mathrm{L2WrB}^{(\mathrm{max})}_{\mathrm{p(Clk\cdot XeCore)}}$$

row 90 - L2_READ_B_XECORE_CLK_PCT

$$\eta_{\mathrm{L2RdB/p(Clk\cdot XeCore)}} = \frac{\mathrm{L2RdB}_{\mathrm{p(Clk\cdot XeCore)}}}{\mathrm{L2RdB}^{(\mathrm{max})}_{\mathrm{p(Clk\cdot XeCore)}}}$$

row 91 - L2_WRITE_B_XECORE_CLK_PCT

$$\eta_{\mathrm{L2WrB/p(Clk\cdot XeCore)}} = \frac{\mathrm{L2WrB}_{\mathrm{p(Clk\cdot XeCore)}}}{\mathrm{L2WrB}^{(\mathrm{max})}_{\mathrm{p(Clk\cdot XeCore)}}}$$

row 92 - L2_READ_WRITE_B_XECORE_CLK_PCT

$$\eta_{\mathrm{L2RdWrB/p(Clk\cdot XeCore)}} = \frac{\mathrm{L2RdWrB}_{\mathrm{p(Clk\cdot XeCore)}}}{\mathrm{L2RdWrB}^{(\mathrm{max})}_{\mathrm{p(Clk\cdot XeCore)}}}$$

row 93 - L1_READ_B_EU_CLK_PCT

$$\eta_{\mathrm{L1RdB/p(Clk\cdot EU)}} = \frac{\mathrm{L1RdB}_{\mathrm{p(Clk\cdot EU)}}}{\mathrm{L1RdBW}^{\mathrm{max}}_{\mathrm{BpClk\cdot EU}}}$$

row 94 - L1_WRITE_B_EU_CLK_PCT

$$\eta_{\mathrm{L1WrB/p(Clk \cdot EU)}} = \frac{\mathrm{L1WrB_{p(Clk \cdot EU)}}}{\mathrm{L1WrBW^{max}_{BpClk \cdot EU}}}$$

row 106 - L2_HIT_RATE_ASSUMED_RANDOM_ACCESS_WITHIN_THE_WORKING_DATA_SET_PCT

$$\mathrm{L2HitRate_{random\ WS\ access}} = \min\left(\frac{|\mathrm{L2Bytes}|}{\mathrm{TotalRequiredL2Size^{1\ instance,\ 1\ wave}_{matB\ alwayshit}}}, 1\right)$$

row 107 - L2_MISS_RATE_PCT

$$\mathrm{L2MissRate} = 1 - \mathrm{L2HitRate_{random\ WS\ access}}$$

row 108 - TOTAL_L2_READ_TRAFFIC_B

$$\mathrm{L2RdB_{total}} = \mathrm{L2Rd^{(total)}_{B}}$$

row 110 - PROBABILITY_OF_MATA_HIT_IN_L2_DURING_A_NON_FIRST_WAVE_PCT

$$\mathrm{r1stwave} = \mathrm{IF}(\mathrm{Size^{(matA)}_{B}} + \mathrm{Size^{(matB)}_{B}} + \mathrm{Size^{(matC,matD)}_{matB}} \leq |\mathrm{L2Bytes}|, 1.0, \mathrm{IF}(\mathrm{K_d im} > 2 \times |\mathrm{WorkingSet}|^{\mathrm{(KinL2)}}_{\mathrm{20Kclksofthreaddivergence}}, 0.0, 1 - \frac{\mathrm{K_d im} - |\mathrm{WorkingSet}|^{\mathrm{(KinL2)}}_{\mathrm{20Kclksofthreadd}}}{|\mathrm{WorkingSet}|^{\mathrm{(KinL2)}}_{\mathrm{20Kclksofthreaddiverg}}}$$

row 111 - PROBABILITY_OF_MATB_HIT_IN_L2_DURING_A_NON_FIRST_WAVE_PCT

$$\mathrm{P^{(matB)}_{L2Hit,after1stwave}} = \mathrm{IF}(\mathrm{Size^{(matA)}_{B}} + \mathrm{Size^{(matB)}_{B}} + \mathrm{Size^{(matC,matD)}_{matB}} \leq |\mathrm{L2Bytes}|, 1.0, 0.0)$$

row 112 - PROBABILITY_OF_MATA_MISS_IN_L2_DURING_A_NON_FIRST_WAVE_PCT

$$\mathrm{P^{(matA)}_{L2Miss,after1stwave}} = 1 - \mathrm{P^{(matA)}_{L2Hit,after1stwave}}$$

row 113 - PROBABILITY_OF_MATB_MISS_IN_L2_DURING_A_NON_FIRST_WAVE_PCT

$$\mathrm{P^{(matB)}_{L2Miss,after1stwave}} = 1 - \mathrm{P^{(matB)}_{L2Hit,after1stwave}}$$

row 116 - TOTAL_HBM_READ_B_AFTER_A_COMPLETION_OF_A_WAVE_CONSIDER_COLD_CACHE

$$\frac{\text{ze}_{\text{B}}^{(\text{matB})} + \text{Size}_{\text{B}}^{(\text{matB})} \times \left(\text{CEILING}(\frac{\text{M}_{\text{d}}\text{im}}{\text{H}_{\text{element}}^{(\text{XeCUTile})}}, 1) - 1\right) \times \text{P}_{\text{L2Miss,after1stwave}}^{(\text{matB})} + \left(\text{L2RdB}_{\text{total}} - \left(\text{Size}_{\text{B}}^{(\text{matA})} + \text{Size}_{\text{B}}^{(\text{matA})} \times \left(\text{CEILING}(\frac{\text{N}_{\text{d}}\text{im}}{\text{W}_{\text{element}}^{(\text{XeCUTile})}}, 1) - 1\right) \times \text{P}_{\text{L2Miss,after1s}}^{(\text{matA})}}{2}$$

row 117 - TOTAL_HBM_WRITE_B

$$\text{MemWrB}^{(\text{total})} = \text{Size}_{\text{matB}}^{(\text{matC,matD})}$$

row 118 - TOTAL_HBM_B

$$\text{MemRdWrB}^{(\text{total})} = \text{MemWrB}^{(\text{total})} + \text{MemRdB}_{\text{coldstart}}^{(\text{wave})}$$

row 121 - HBM_READ_B_CLK

$$\text{MemRdB}_{\text{pClk}} = \frac{\text{MemRdB}_{\text{coldstart}}^{(\text{wave})}}{\text{T}_{\text{clk}}^{(\text{total})}}$$

row 122 - HBM_WRITE_B_CLK

$$\text{MemWrB}_{\text{pClk}} = \frac{\text{MemWrB}^{(\text{total})}}{\text{T}_{\text{clk}}^{(\text{total})}}$$

row 123 - HBM_TOTAL_B_CLK

$$\text{MemRdWrB}_{\text{pClk}} = \frac{\text{MemRdWrB}^{(\text{total})}}{\text{T}_{\text{clk}}^{(\text{total})}}$$

row 126 - HBM_BW_PCT

$$\eta_{\text{MemBW}} = \frac{\text{MemRdWrB}_{\text{pClk}}}{\text{MemBW}_{\text{BpClk}}^{\text{max}}}$$

row 127 - GTI_READ_BW_PCT

$$\eta_{\mathrm{GTIRdBW}} = \frac{\mathrm{MemRdB}_{\mathrm{pClk}}}{\mathrm{GtiRdBW}^{\mathrm{max}}_{\mathrm{BpClk}}}$$

row 128 - GTI_WRITE_BW_PCT

$$\eta_{\mathrm{GTIWrBW}} = \frac{\mathrm{MemWrB}_{\mathrm{pClk}}}{\mathrm{GtiWrBW}^{\mathrm{max}}_{\mathrm{BpClk}}}$$